

ODD-CYCLE COLOURINGS OF ABELIAN CAYLEY GRAPHS COME FROM CHARACTERS

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ABSTRACT. We show that a Cayley graph of an abelian group has a homomorphism to the odd cycle C_{2k+1} if and only if some character maps every generator into the closed arc $[k/(2k+1), (k+1)/(2k+1)]$ of $\mathbb{R}/\mathbb{Z}$. In particular such a graph is 3-colourable if and only if some character maps every generator into $[1/3, 2/3]$. The proof averages the discrete winding number of Krebs and Sankar over the group with an invariant mean. Consequences: Payan’s theorem that a cube-like graph is never 3-chromatic, and the exponent-4 case of the theorem of Krebs and Sankar, follow at once; a distance graph $G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; and for a number field F the plane F^2 is 3-colourable if and only if a character of the group generated by the unit vectors maps them into $[1/3, 2/3]$. It also follows that in every abelian group, every set of Bohr recurrence is a set of 3-chromatic recurrence: the case of three colours of a question of Katznelson. For $F = \mathbb{Q}(\sqrt{d})$ and finitely many unit vectors the last condition is a question on a 4-dimensional torus, which exact branch-and-bound certificates settle. We obtain $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$, $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$ and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$: five fields for which searches for finite non-3-colourable unit-distance graphs had failed.

1. INTRODUCTION

For an abelian group Γ and a subset $S = -S$ of $\Gamma \setminus \{0\}$, the Cayley graph $\text{Cay}(\Gamma, S)$ has vertex set Γ and an edge between g and $g + s$ for $g \in \Gamma$, $s \in S$. A *character* is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$. For $x \in \mathbb{R}/\mathbb{Z}$ let $\|x\|$ be its distance to 0.

Theorem 1. *Let $k \geq 1$ and $n = 2k + 1$. The graph $\text{Cay}(\Gamma, S)$ has a homomorphism to the cycle C_n if and only if there is a character ξ of Γ with $\xi(s) \in [k/n, (k+1)/n]$ for every $s \in S$. In particular, $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps every element of S into $[1/3, 2/3]$.*

A homomorphism to C_{2k+1} is a circular $(2k+1)/k$ -colouring. Let $\kappa(S) = \sup_{\xi} \inf_{s \in S} \|\xi(s)\|$; the supremum is attained, as the dual group is compact and $\xi \mapsto \inf_s \|\xi(s)\|$ is upper semicontinuous. The theorem says

$$\chi_c(\text{Cay}(\Gamma, S)) \leq 2 + \frac{1}{k} \iff \kappa(S) \geq \frac{k}{2k+1} \quad (k \geq 1).$$

The *if* direction is the classical inequality $\chi_c \leq 1/\kappa$: for distance graphs it is Zhu’s multiplier method, surveyed by Liu [18, §2], and for Cayley graphs of $\mathbb{Z}/p$ it appears in [1, Theorem 3.2]. The content of the theorem is the converse at the thresholds $2 + 1/k$, among them $\chi \leq 3$.

Steps 1 and 2 of the proof are the discrete winding number of Krebs and Sankar [17] (their Definition 3.5, Propositions 3.3 and 3.6, Remark 3.7, and the parity argument in the proof of their Theorem 4.1), which they use to show that groups of exponent 4 have no 3-chromatic Cayley graphs. What is new, as far as we know, is Step 3, which averages the winding over the group, and the characterisation that results (Section 7).

For $k = 1$ the theorem has a reading in terms of recurrence. A set $S \subseteq \Gamma$ is a set of *r-chromatic recurrence* if every r -colouring of Γ has two points of one colour that differ by an element of S ,

Date: Draft, 3 October 2026.

2020 Mathematics Subject Classification. Primary 05C15; Secondary 05C25, 11R11, 52C10.

Key words and phrases. Cayley graph, abelian group, 3-colouring, circular colouring, distance graph, unit-distance graph, Hadwiger–Nelson problem.

These results were obtained with the assistance of an AI system (Claude, by Anthropic), which carried out the computations and the checks and helped to draft this note. A separate AI agent refereed the draft, with its own checker and tests, and its findings are incorporated. No mathematician has checked it yet. The author is not a professional mathematician and submits it for independent verification.

and a set of *Bohr recurrence* if it meets every Bohr neighbourhood of 0. Katznelson's question [15, 12] asks whether every set of Bohr recurrence is a set of r -chromatic recurrence for every r ; it is open, and the answer is not known for any countably infinite abelian group [12]. For $r = 2$ the answer is easily yes. Theorem 1 gives $r = 3$:

Corollary 2. *A set $S \subseteq \Gamma \setminus \{0\}$ is a set of 3-chromatic recurrence if and only if it meets the set $\{g : \|\xi(g)\| < 1/3\}$ for every character ξ . In particular, every set of Bohr recurrence is a set of 3-chromatic recurrence, and every 3-colourable abelian Cayley graph has a 3-colouring whose colour classes are $\xi^{-1}[0, 1/3)$, $\xi^{-1}[1/3, 2/3)$ and $\xi^{-1}[2/3, 1)$ for a single character ξ .*

Proof. Apply Theorem 1 with $k = 1$ to $S \cup (-S)$; the colouring is the one in the *if* direction of the proof. $\square$

We have not found the case $r = 3$ stated in [12, 13, 19]; we have not been able to read [15].

Section 2 proves the theorem. Section 3 derives Payan's theorem [21] and the exponent-4 case of Krebs and Sankar [17]. Section 4 treats distance graphs. Section 5 applies the theorem to the unit-distance graph of the plane over a number field, the setting that led to it, and Section 6 records that nothing of the kind holds for four colours.

2. PROOF OF THEOREM 1

If. Let $c(g) = 2\lfloor n\xi(g) \rfloor \bmod n$, where $\xi(g)$ is read in $[0, 1)$. Along an edge, $n\xi$ moves by an amount in $[k, k+1]$ modulo n , so $\lfloor n\xi \rfloor$ moves by k or $k+1$ modulo n , and c moves by $2k \equiv -1$ or $2k+2 \equiv 1$. So c is a homomorphism to C_n , whose vertices are $\mathbb{Z}/n$ with $j \sim j \pm 1$.

Only if. Let $c : \Gamma \rightarrow \mathbb{Z}/n$ be a homomorphism to C_n , and let Γ' be the subgroup generated by S . A character of Γ' extends to Γ , since $\mathbb{R}/\mathbb{Z}$ is divisible, so it suffices to find the character on Γ' . We work in Γ' because $\text{Cay}(\Gamma', S)$ is connected, which Step 2 uses: averaging over all of Γ would fail when $\text{Cay}(\Gamma, S)$ is disconnected, since different components can wind in opposite directions. For $g \in \Gamma'$ and $s \in S$ let $\sigma(g, s) \in \{1, -1\}$ be the lift of $c(g+s) - c(g)$; then $\sigma(g+s, -s) = -\sigma(g, s)$.

Step 1: squares do not wind [17, Proposition 3.6]. For $s, t \in S$ the two integers $\sigma(g, s) + \sigma(g+s, t)$ and $\sigma(g, t) + \sigma(g+t, s)$ lie in $\{-2, 0, 2\}$ and are congruent to $c(g+s+t) - c(g)$ modulo n . Two distinct elements of $\{-2, 0, 2\}$ differ by 2 or 4, which n (odd, at least 3) does not divide; so they are equal.

Step 2: the winding of a closed walk [17, Definition 3.5, Propositions 3.3 and 3.6, Remark 3.7]. Let Z be the abelian group generated by symbols $[s]$, $s \in S$, subject to $[-s] = -[s]$. It is free on one element of each pair $\{s, -s\}$, except that $2[s] = 0$ when $2s = 0$; so the parity of the number of symbols is a well-defined homomorphism $Z \rightarrow \mathbb{Z}/2$. Let R be the kernel of $Z \rightarrow \Gamma'$, $[s] \mapsto s$.

A closed walk from g with steps $s_1, \dots, s_N$ ($\sum s_j = 0$) has the sum $\sum_j \sigma(g+s_1+\dots+s_{j-1}, s_j) \equiv 0 \pmod{n}$. By Step 1 this sum does not change when two consecutive steps are exchanged, nor when a step s immediately followed by $-s$ is deleted. Every walk can be brought to a normal form by these moves (sort the steps by exchanges, then cancel the pairs $s, -s$), so the sum depends only on the base point and on the class $r = \sum_j [s_j] \in R$. It does not depend on the base point either: going from $g+s$ to g , doing the walk and coming back gives the same sum, and $\text{Cay}(\Gamma', S)$ is connected. Write the sum as $nw_c(r)$. Then $w_c : R \rightarrow \mathbb{Z}$ is a homomorphism (concatenate walks). Since $nw_c(r)$ is a sum of N odd numbers and n is odd, $w_c(r)$ has the parity of r .

Step 3: averaging. As Γ' is abelian, it is amenable: there is a mean m on the bounded functions on Γ' that is invariant under translations. For finite Γ' it is the plain average, and for finitely generated Γ' a limit along Følner sets. Let

$$f(s) = m(g \mapsto \sigma(g, s)) \in [-1, 1].$$

Then $f(-s) = -f(s)$. For a closed walk $(s_1, \dots, s_N)$ that represents $r \in R$, the function $g \mapsto \sum_j \sigma(g+s_1+\dots+s_{j-1}, s_j)$ is the constant $nw_c(r)$ by Step 2. Applying m , which is linear

and invariant under translations, gives

$$\sum_{j=1}^N f(s_j) = n w_c(r).$$

Step 4: the character. Put $\xi(s) = \frac{1}{2}(f(s)/n + 1) \in [k/n, (k+1)/n]$; then $\xi(-s) = 1 - \xi(s)$. If $2s = 0$, then $\sigma(g + s, s) = -\sigma(g, s)$ for all g , so $f(s) = 0$ and $\xi(s) = 1/2$. Hence ξ is well defined modulo 1 on Z . For $r \in R$ represented by a walk of length N , $\sum_j \xi(s_j) = \frac{1}{2}(w_c(r) + N)$, an integer by the parity in Step 2. So ξ vanishes on R modulo 1 and defines a character of $Z/R = \Gamma'$, with $\xi(S) \subseteq [k/n, (k+1)/n]$. $\square$

Remark 3. For $\Gamma = \mathbb{Z}^2$ with the standard generators this is the height-function picture of 3-colourings of the square lattice, in which the height changes by ± 1 along edges and by 0 around squares (compare [23]); f is the averaged slope. Theorem 1 says that the averaged slopes of any colouring of any abelian Cayley graph form a character as above. The colouring c itself need not come from a character: for $\Gamma = \mathbb{Z}^2$ most 3-colourings do not.

3. GROUPS OF SMALL EXPONENT

Corollary 4. *Let Γ be an abelian group of exponent 2 or 4. Then no Cayley graph of Γ has chromatic number 3.*

Proof. A character of Γ takes values in $\frac{1}{4}\mathbb{Z}/\mathbb{Z}$, and the only such value in $[1/3, 2/3]$ is $1/2$. So if $\text{Cay}(\Gamma, S)$ is 3-colourable, a character maps S to $1/2$, and $g \mapsto \lfloor 2\xi(g) \rfloor$ is a proper 2-colouring. $\square$

For exponent 2 this is Payan's theorem on cube-like graphs [21], which Cervantes and Krebs reproved [4]. For exponent 4 it is a case of the theorem of Krebs and Sankar [17, Theorem 1.2]: an abelian group has a 3-chromatic Cayley graph if and only if its exponent is not 1, 2 or 4. Their proof uses the same winding number: for exponent 4 they show that the discrete fundamental group is torsion, and by their Theorem 4.1 a non-bipartite graph with a torsion fundamental group is not 3-colourable. The same argument as for Corollary 4 gives, for instance, that a Cayley graph of an abelian group of exponent 8 that maps to C_5 is bipartite, since $1/2$ is the only element of $\frac{1}{8}\mathbb{Z}/\mathbb{Z}$ in $[2/5, 3/5]$. For C_3 the values $3/8$ and $5/8$ are also available, and indeed such groups have 3-chromatic Cayley graphs [17].

4. DISTANCE GRAPHS

For a finite set D of positive integers, the distance graph $G(\mathbb{Z}, D)$ is $\text{Cay}(\mathbb{Z}, \pm D)$. Its characters are $x \mapsto \alpha x$, $\alpha \in \mathbb{R}/\mathbb{Z}$, and $\kappa(D) = \max_{\alpha} \min_{d \in D} \|\alpha d\|$ is the parameter of the lonely runner problem [18, 22].

Corollary 5. *$G(\mathbb{Z}, D)$ is 3-colourable if and only if $\kappa(D) \geq 1/3$; it maps to C_{2k+1} if and only if $\kappa(D) \geq k/(2k+1)$.*

The set of α with $\|\alpha d\| \geq 1/3$ for all $d \in D$ is a finite union of closed intervals whose endpoints are among the points $(j+1/3)/d$ and $(j+2/3)/d$. So it is non-empty if and only if it contains one of these points, and the test is finite. We compared it, for all 1 747 sets $D \subseteq [1, 24]$ with $|D| = 3$ and $\gcd(D) = 1$, with 3-colourability of the segment $[0, 12 \max D + 59]$ decided by a SAT solver. They agree, and the 74 sets that need four colours are exactly those of Zhu's classification [24] of 3-element sets with $\chi(G(\mathbb{Z}, D)) = 4$, namely $\{1, 2, 3m\}$ and $\{a, b, a+b\}$ with $a \not\equiv b \pmod{3}$ (after division by the gcd). Liu's survey asks whether $\chi_c(G(\mathbb{Z}, D)) < 1/\kappa(D)$ can happen when $|D| = 3$ [18, Problem 3]. By Corollary 5, if one of the two numbers is at most 3, then both equal 2 or both lie in the same interval $(2 + 1/(k+1), 2 + 1/k]$.

5. PLANES OVER NUMBER FIELDS

Let F be a field of characteristic 0 with $i \notin F$, and $L = F(i)$. Identify F^2 with L by $(x, y) \mapsto x + iy$; the unit vectors are $T = \{z \in L : z\bar{z} = 1\}$, and the unit-distance graph of F^2 is $\text{Cay}(L, T)$, a disjoint union of copies of $\text{Cay}(A, T)$ with $A = \mathbb{Z}[T]$.

Corollary 6. $\chi(F^2) \leq 3$ if and only if some character of A maps every unit vector into $[1/3, 2/3]$. If for a finite set $U \subset T$ no character of the group $\mathbb{Z}U$ maps U into $[1/3, 2/3]$, then $\chi(F^2) \geq 4$, and some finite unit-distance graph over F is not 3-colourable.

The last assertion is the theorem of de Bruijn and Erdős [6]. If F is a number field, then $\mathbb{Z}U$ is a finitely generated subgroup of the $\mathbb{Q}$ -vector space L , hence free of rank at most $[L : \mathbb{Q}]$; for a quadratic field F the test for a finite U is a question on a torus of dimension at most 4. A locally constant 3-colouring at a place of F [7, 20, 9] is a colouring of a finite quotient of A , and Theorem 1 applied to that quotient gives such a character.

5.1. Real quadratic fields. For $F = \mathbb{Q}(\sqrt{d})$ with $d > 1$ squarefree, $\chi(F^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [14, 7, 20], $\chi(F^2) \leq 3$ if moreover $d \not\equiv 2 \pmod{3}$ [7, 20], and $\chi(F^2) \leq 4$ if $d \equiv 3 \pmod{8}$ [7, Theorem 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [20]. In [10] we found finite non-3-colourable unit-distance graphs over $\mathbb{Q}(\sqrt{d})$ for 27 values $d \equiv 11 \pmod{12}$ below 1000, but searches for such graphs failed for $d = 83, 107, 143, 167, 203$, the values that the local–global question of [9] predicted to need four colours and that remained open below 210.

Fix a denominator D and let U_D be the unit vectors $((a + b\sqrt{d})/D, (c + e\sqrt{d})/D)$, $a, b, c, e \in \mathbb{Z}$, one of each pair $\pm u$. A character with $\xi(U_D) \subseteq [1/3, 2/3]$ gives $f \in [1/3, 2/3]^{U_D}$ (f_u the lift of $\xi(u)$ in $[1/3, 2/3]$) with $\langle r, f \rangle \in \mathbb{Z}$ for every integer relation r among the vectors of U_D . Using an LLL-reduced basis of the relations, we rule this out by branch and bound over the integers $\langle r, f \rangle$: each leaf carries a rational vector y (a Farkas certificate) such that $\langle y, \cdot \rangle$ applied to the equations fixed at that leaf cannot hold on the box $[1/3, 2/3]^{U_D}$. A short program checks a certificate in exact integer and rational arithmetic: the vectors have length 1 ($a^2 + db^2 + c^2 + de^2 = D^2$, $ab + ce = 0$), the relations hold, every branch covers every integer value in its range, and every Farkas vector is valid. Any set of valid relations gives necessary conditions, so infeasibility is a proof whether or not the relations form a basis.

Theorem 7. $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for $d = 83, 107, 203$; $4 \leq \chi(\mathbb{Q}(\sqrt{143})^2) \leq 5$; and $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$.

Proof. The lower bounds are Corollary 6 with the certificates of Table 1 [8]. The upper bound 4 holds because $83 \equiv 107 \equiv 203 \equiv 3 \pmod{8}$ [7, Theorem 10]; see also [2, Theorem 2.2]. For $d = 143$ the place above 11 is ramified in $\mathbb{Q}(\sqrt{143})$ with residue field $\mathbb{F}_{11}$, and i is not in its completion; unit vectors are integral there and reduce to the circle $x^2 + y^2 = 1$ of $\mathbb{F}_{11}^2$, whose graph is 5-colourable [10], which colours each coset of the valuation ring. $\square$

d	D	vectors	relations	nodes	leaves
83	510	54	50	6 886	4 664
107	1170	90	86	12 785	9 618
143	1740	54	50	3 861	2 710
167	1560	54	50	8 552	6 094
203	1530	78	74	1 072	732

TABLE 1. Certificates that no character of $\mathbb{Z}U_D$ maps U_D into $[1/3, 2/3]$ (vectors counted up to sign). In each case the vectors span a group of rank 4 and the relations listed generate all relations.

With the values of [10], $\chi(\mathbb{Q}(\sqrt{d})^2)$ is now known for every squarefree $d < 143$ except $d = 47$. $\mathbb{Q}(\sqrt{167})$ is the first admissible field of [9]: no place has a locally constant 4-colouring, so $\chi(\mathbb{Q}(\sqrt{167})^2) \geq 4$ is also the three-colour case of the local–global question there. For $d = 203$ the result shows that no level of the place above 7 is 3-colourable, which [9] had checked only up to level 3. For $d = 83$ and $D = 510$, colouring-guided growth had stopped at 6 008 points because a 3-colouring of the ball of radius 2 extended to all 215 478 candidate points, so the obstruction does not lie in a small ball. A subset of 28 of the 54 vectors, found by greedy deletion, already suffices for $d = 83$. For $\mathbb{Q}(\sqrt{11})$, a subset of 16 vectors with $D = 30$ suffices, in agreement with the 76-vertex graph of [10].

The lower bounds rest on Theorem 1, whose averaging step is not constructive, and on the certificates; we give no explicit finite graph for the five fields. We expect such graphs to be large, which would explain why the searches failed, but this is speculation: Step 3 says nothing about their size. A quantitative version of Step 3 over Følner boxes, or an explicit certified graph, would remove the dependence on the infinite averaging.

All five denominators are even, as they must be. For odd D every vector of U_D has $a + b + c + e$ odd, because $a^2 + db^2 + c^2 + de^2 = D^2$ and d is odd; so $x \mapsto \frac{1}{2}(a + b + c + e)$, for $Dx = (a + b\sqrt{d}) + i(c + e\sqrt{d})$, is a character of $\mathbb{Z}U_D$ with the value $1/2$ on U_D , and $\text{Cay}(\mathbb{Z}U_D, U_D)$ is bipartite. The infeasible denominators we found are all multiples of 6, but some multiples of 6 are feasible (for instance $D = 1170$ for $d = 203$). As controls, the test is feasible for 40 pairs (d, D) with $d \in \{2, 3, 5, 6, 7, 15, 19, 31, 39, 43\}$, whose planes are known to be 3-colourable, and for the even denominators $(d, D) = (107, 1450)$, $(143, 1752)$, $(167, 1820)$ and $(203, 1590)$, whose unit vectors also span a group of rank 4.

Question 8. Is $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every squarefree $d \equiv 11 \pmod{12}$? In terms of Corollary 6 and [9]: for real quadratic fields, does a character that maps all unit vectors into $[1/3, 2/3]$ exist only when a place gives a locally constant 3-colouring?

6. FOUR COLOURS

Step 1 fails for K_4 , and so does the theorem: $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ is 4-colourable, but every character of $(\mathbb{Z}/2)^2$ vanishes on some nonzero element, so none maps the three generators into $[1/4, 3/4]$. The inequality $\chi_c \leq 1/\kappa$ holds for all k , so a character with $\xi(S) \subseteq [1/4, 3/4]$ still gives a 4-colouring. For the 108 unit vectors of $\mathbb{Q}(\sqrt{47})$ with denominator 240, which underlie the search for a five-chromatic triangle-free unit-distance graph in [9], a floating-point computation finds none; we have not certified this.

7. LITERATURE

The closest work is that of Krebs and Sankar [17], whose winding number is Steps 1 and 2; they use it for torsion fundamental groups, without characters or averaging. The *if* direction of Theorem 1 is classical [18, 1, 11]. The following are consistent with Theorem 1 and do not imply it: Cervantes and Krebs [3, 4, 5] (in [5], 3-colourability for small dimension and rank is characterised by two forbidden subgraphs, diamond lanyards and $C_{13}(1, 5)$, and $C_{13}(1, 5)$ indeed has no suitable character); Krebs and Leyva [16]; the work on Katznelson’s question on Bohr recurrence [12, 13, 19], where we found no statement for three colours (we have not seen [15] itself); and Perarnau and Serra [22, §3.5], which has $\chi_f \leq \chi_c \leq 1/\kappa$. We did not find Theorem 1 in these works, in the literature on distance graphs and the lonely runner problem through Liu’s survey [18], or in web searches on characters, circular colourings and odd-cycle homomorphisms of Cayley graphs. The proof uses only classical tools, so an earlier occurrence is possible.

By Theorem 1, whether a finite abelian Cayley graph, given by Γ and S , maps to C_{2k+1} can be decided by checking the $|\Gamma|$ characters, with $O(|\Gamma| |S|)$ operations: polynomial in the size of the graph.

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